

Immersive Interaction and Interpretation of Ancient Spatial and Shape Data: XR for Archaeological Research

Peter J. Cobb

School of Humanities, Faculty of Arts
University of Hong Kong
Hong Kong SAR, China
pcobb@hku.hk

Hayk Azizbekyan

School of Humanities, Faculty of Arts
University of Hong Kong
Hong Kong SAR, China
hayk@connect.hku.hk

Abstract

We will demonstrate our innovations for data collection and interaction during archaeological fieldwork using mixed and augmented reality. Archaeologists understand past human life by researching the objects and spaces that were created and modified during past human activity. Thus, our research is inherently three-dimensional (3D) in the real world, and our data includes the spatial information from our sites, such as architectural remains, and the shape information of the artifacts we uncover. Although 3D graphics have provided new ways to precisely capture archaeological data over the last decade [Roosevelt et al., 2015], the use of these data is limited by our 2D interfaces. Mixed and augmented reality headsets have the significant potential to revolutionize the ways that archaeologists conduct their fieldwork by changing how they interact with and interpret their data in the field. Our experiments with eXtended Reality devices during our archaeological fieldwork demonstrate the way forward for archaeologists working all over the world.

CCS Concepts

• **Computing methodologies** → Computer graphics; Graphics systems and interfaces; Mixed / augmented reality; • **Information systems** → Information systems applications; Spatial-temporal systems; • **Applied computing** → Arts and humanities.

Keywords

Archaeology, Mixed Reality, Augmented Reality, Immersive Data Interaction, Digital Humanities

ACM Reference Format:

Peter J. Cobb and Hayk Azizbekyan. 2025. Immersive Interaction and Interpretation of Ancient Spatial and Shape Data: XR for Archaeological Research. In *SIGGRAPH Asia 2025 XR (SA XR '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3761667.3761956>

1 Introduction

Archaeology is the study of the human past based on the materials left behind by past societies – from the architectural remains of



Figure 1: Using the Microsoft HoloLens 2 during archaeological fieldwork in Armenia.

houses and other buildings, to the pottery, seeds, and animal bones from meals, and beyond. To uncover these data, archaeologists conduct excavations at sites all over the world. Digging uncovers the artifacts and features, like walls and hearths, in their positions of last use and deposition, enabling the understanding of how a site and objects were used in the past. The primary goal of archaeological fieldwork, therefore, is to precisely record the locations and spatial relationships among these findings.

Today, many archaeologists use photogrammetry and other 3D data capture techniques while they excavate to record spatial information, but these 3D models are often only used later for simple measurements and to extract 2D illustrations. We have identified that it would be more useful to deploy those data directly during excavation to help with interpretation and excavation decision-making, particularly around studying spatial relationships at an archaeological site [Dilena and Soressi, 2020]. We experimented with several mixed reality headsets in the field to support this type of data interaction [Cobb and Azizbekyan, 2024].

Additionally, we also recognized the potential for mixed and augmented reality headsets to improve how archaeologists dig and collect data in the field during excavation. In order to more clearly see ancient remains, archaeologists often intentionally cut the dirt to flat planes. Given that mixed reality headsets can scan their surroundings and precisely project geometry into the real world, we recognized that virtual planes could be used to guide archaeological digging. Finally, since archaeologists need their hands free to move dirt and touch artifacts, we see significant potential for

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA XR '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2168-7/2025/12

<https://doi.org/10.1145/3761667.3761956>

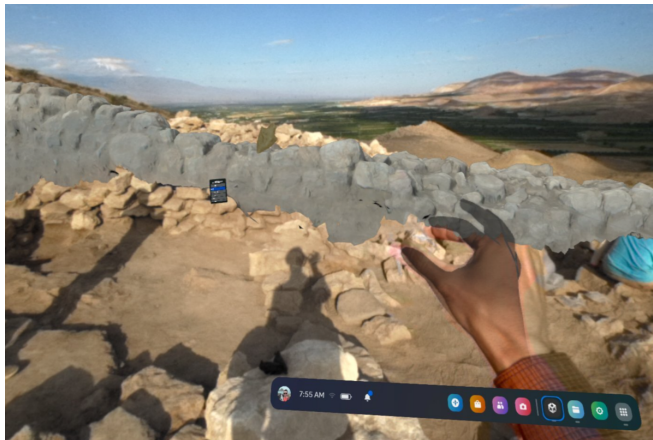


Figure 2: Placing the 3D model of a previously removed stone wall in the Quest's mixed reality mode.

using even 2D augmented reality smartglasses, with voice recognition, to gather and display basic data during excavation, such as photographs and textual information.

2 Demonstrations

We have developed several innovative workflows using a variety of eXtended Reality headsets, and we have deployed these during an active archaeological fieldwork project in the South Caucasus country of Armenia [Cobb et al., 2024]. Each workflow addresses one of the opportunities for technological improvement of excavation, as discussed above.

2.1 Viewing Removed Data with the Quest

Archaeology is a destructive science – in order to uncover new data, we must remove the dirt and features in the ground. Yet, we often want to remember precisely where we uncovered something that we previously excavated as we dug to the deeper levels. We capture 3D models that are geolocated after removing each volume of dirt or other materials, a volume we call a ‘context.’ Using mixed reality, we now have the opportunity to compare these removed contexts to the ones we are currently digging [Figure 2]. We developed a custom app in the Quest devices that enables us to recall any of our 3D models of context volumes that we have previously excavated and place them into the visual field of the user, at their original locations within the excavation trench.

2.2 Excavating Flat Surfaces with the HoloLens

One main skill developed over time by archaeologists is the ability to trowel the dirt to a flat surface. This enables us to see remains more clearly, including lines that divide different colors of dirt. To speed up the process of training archaeologists and even help experienced archaeologists, we developed a custom application for the HoloLens 2 that displays a virtual flat plane in the environment. The archaeologist sets the depth of the plane, and then can dig directly to that depth, helping the excavator shape a precise flat surface at a constant depth. The HoloLens 2 is continuously 3D

scanning the environment, so immediately when that depth is reached, the plane appears.

2.3 Collecting Data with Custom Smartglasses

Currently, archaeologists who have digital workflows usually record basic data on a tablet or smartphone. This presents challenges during fieldwork, since the device must be used with hands covered by gloves and dirt, and the device must be removed from a pocket periodically. The heat of the sun and the dirt can also impact devices and their batteries. We tested a few commercial augmented reality smartglasses to attempt to replace our smartphones for basic data collection. We collect information such as the date and description of each context we excavate, plus we take many 2D photographs to document our progress. We also write a narrative journal in Evernote to record our activities. The commercial smartglasses we tried were not sophisticated enough to enable us to replace our smartphones, so now we have built our own custom smartglasses to improve data collection. We are currently testing these custom smartglasses, which have a screen, microphone, and a camera, at our site in Armenia.

3 Conclusion and Future Work

Our work has already had several successes in demonstrating how mixed and augmented reality headsets can greatly enhance archaeological fieldwork. At the same time, we have recognized the limitations of the current headset hardware options, that are often bulky, with short battery lives, and may not have screens that are visible in bright sunlight. We already created our own custom smartglasses to overcome some of these hurdles, and we are continuously looking to form partnerships with technologists to build better devices.

Acknowledgments

We thank the Tam Wing Fan Innovation Wing of the Faculty of Engineering and the Libraries at the University of Hong Kong for equipment and support. Funding support for our archaeological and technological research has also been provided by a grant from the Research Grants Council (RGC) of the Hong Kong Special Administrative Region (HKSAR), China (project no. HKU 17607022).

References

- Peter J. Cobb and Hayk Azizbekyan. 2024. Experiments with Mixed and Augmented Reality (MR/AR) for Archaeological Data Collection and Use During Fieldwork: Vision for the Future. *Journal of Computer Applications in Archaeology* 7, 1, 370–387. DOI: <https://doi.org/10.5334/jcaa.140>
- Peter J. Cobb, Elvan Cobb, Hayk Azizbekyan, Artur Petrosyan, and Boris Gasparyan. 2024. Defending the Vedi River Valley of Armenia: The Fortification and Refortification of a Flatland-Mountain Interface. *Bulletin of the American Society of Overseas Research* 392, 151–77. DOI: <https://doi.org/10.1086/732090>
- Miguel A. Dilena and Marie Soressi. 2020. Reconstructive Archaeology: In Situ Visualization of Previously Excavated Finds and Features through an Ongoing Mixed Reality Process. *Applied Sciences* 10, 21, 7803. DOI: <https://doi.org/10.3390/app10217803>
- Christopher H. Roosevelt, Peter Cobb, Emanuel Moss, Brandon R. Olson, and Sinan Ünüsoy. 2015. Excavation is <strike>Destruction</strike> Digitization: Advances in Archaeological Practice. *Journal of Field Archaeology* 40, 3, 325–346. DOI: <https://doi.org/10.1179/2042458215Y.0000000004>